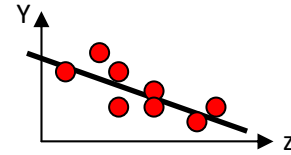


Overview of linear regression models and terminology

Simple linear regression

Model $Y = \beta_0 + \beta_1 z + \varepsilon$
Fit of 1-dimensional regression line
to 2-dimensional observations (z, Y)



Univariate Multiple Linear Regression (UMLR)

Model $Y = \beta_0 + \beta_1 z_1 + \beta_2 z_2 + \dots + \beta_r z_r + \varepsilon = [1 \ z_1 \ \dots \ z_r] \begin{bmatrix} \beta_0 \\ \vdots \\ \beta_r \end{bmatrix} + \varepsilon = \mathbf{z}\boldsymbol{\beta} + \varepsilon$

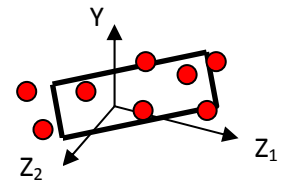
Y = univariate (scalar) response variable (random variable)

$\mathbf{z} = [1 \ z_1 \ \dots \ z_r]$ = vector of predictor variables (deterministic)
(usually continuous, but can also be categorical to include ANOVA-model)

$\boldsymbol{\beta} = [\beta_0 \ \beta_1 \ \dots \ \beta_r]^T$ = vector of unknown regression coefficients (deterministic)

ε = model error (random variable)

Fit of r -dimensional regression hyperplane
to $(r+1)$ -dimensional observations $(\mathbf{z}, Y)$



Multivariate Multiple Linear Regression (MMLR) = General Linear Model (GLM)

Model $[Y_1 \ Y_2 \ \dots \ Y_m] = [1 \ z_1 \ \dots \ z_r] \begin{bmatrix} \beta_{01} & \beta_{02} & \dots & \beta_{0m} \\ \beta_{11} & \beta_{12} & & \beta_{1m} \\ \vdots & \vdots & & \vdots \\ \beta_{r1} & \beta_{r2} & \dots & \beta_{rm} \end{bmatrix} + [\varepsilon_1 \ \varepsilon_2 \ \dots \ \varepsilon_m]$

$\mathbf{Y} = \mathbf{z}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

$\mathbf{Y} = [Y_1 \ Y_2 \ \dots \ Y_m]$ = vector of response variables (multivariate random variable)

$\mathbf{z} = [1 \ z_1 \ \dots \ z_r]$ = vector of common predictor variables (deterministic)
(usually continuous, but can also be categorical to include MANOVA-model)

$\boldsymbol{\beta} = \{\beta\}_{ij}$ = matrix of unknown regression coefficients (deterministic)

$\boldsymbol{\varepsilon} = [\varepsilon_1 \ \varepsilon_2 \ \dots \ \varepsilon_m]$ = vector of model errors (multivariate random variable)

IF $\boldsymbol{\Sigma}_{\varepsilon} = \boldsymbol{\Sigma}_Y = \text{Cov}[\{\varepsilon_1, \dots, \varepsilon_m\}] = \begin{bmatrix} \sigma_{11} & 0 & \dots & 0 \\ 0 & \sigma_{22} & & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \dots & \sigma_{mm} \end{bmatrix}$

THEN the regression equations for $Y_1, Y_2, \dots, Y_m$ can be considered one-at-a-time independent of the others, thus giving an m-multiple UMLR model

ELSE IF $\boldsymbol{\Sigma}_{\varepsilon} = \boldsymbol{\Sigma}_Y = \text{Cov}[\{\varepsilon_1, \dots, \varepsilon_m\}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \dots & \sigma_{1m} \\ \sigma_{21} & \sigma_{22} & & \sigma_{2m} \\ \vdots & \vdots & & \vdots \\ \sigma_{m1} & \sigma_{m2} & \dots & \sigma_{mm} \end{bmatrix}$

THEN a true multivariate MMLR-model is appropriate

The concept of linear regression

More general UMLR-model $Y = \beta_0 f_0(z_1, \dots, z_r) + \beta_1 f_1(z_1, \dots, z_r) + \dots + \beta_r f_r(z_1, \dots, z_r) + \varepsilon$
where $f_0, f_1, \dots, f_r$ are known (assumed) functions of the predictor variables vector

Model is still linear in the unknown regression coefficients $\beta_0, \beta_1, \dots, \beta_r$
 $\Rightarrow$ can be treated as UMLR above by simply substituting $1 \rightarrow f_0(z_1, \dots, z_r)$,
 $z_1 \rightarrow f_1(z_1, \dots, z_r)$, $\dots$, $z_r \rightarrow f_r(z_1, \dots, z_r)$
(same methodology can be used for the other linear regression models)

Most often, however $f_j(z_1, \dots, z_r) = z_j$ (as in standard UMLR) or polynomial in $z_1, \dots, z_r$ (e.g. a term like z_1^2) or in cross-terms (e.g. a term like $z_1 z_2^2$)

Non-linear regression transformable to linear model

Example model $Y = \beta_0 \exp(\beta_1 z)$
Model is non-linear in the regression coefficients β_0, β_1 (and in z)

However, for this example, taking logarithms
 $Y^* \stackrel{\text{def}}{=} \log(Y) = \log(\beta_0) + \beta_1 z \stackrel{\text{def}}{=} \beta_0^* + \beta_1^* z^*$,
simple linear regression can then be performed on transformed observations
 $(z^*, Y^*) = (z, \log(Y))$
followed by back-transformation of regression coefficients
 $\beta_0 = \exp(\beta_0^*)$, $\beta_1 = \beta_1^*$

Non-linear regression NOT transformable to linear regression

Example model $Y = \cos \beta_0 z + \sqrt{\beta_1 z}$
Model is non-linear in the regression coefficients β_0, β_1 (and in z) and cannot in any simple way be transformed to a linear regression problem

Examples like this must be treated numerically using optimization methods

Some important questions for linear regression

Basic Estimation of unknown regression coefficients and variance (or covariance matrix)
Test of overall statistical significance of model
Model control
Quantitative measure of model fit
Inference (hypothesis test, confidence intervals) for estimated model parameters
Prediction of new observations

Further Model simplification procedures
Search for “best model” (optimal subset of predictor variables)
